Mamadou Ben Hamidou CISSOKO Ph.D.

Machine Learning Scientist | Time Series, Generative Models & LLM Systems

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SUMMARY

Machine-learning scientist (Ph.D.) with 6 years of building deep-learning models for messy, real-world data: sparse and irregular ICU records, 20-megapixel drone imagery and tool-using LLM agents. I design new architectures (diffusion, JEPA, state-space and recurrent models) and carry them to tested, reproducible PyTorch code on HPC. Track record: 2 journal and 2 conference papers, 3 more under review, open-source releases, and reviewer for NeurIPS, UAI and AISTATS. Looking for an **applied research scientist, research engineer or ML engineer** role where models have to work on real data.

CORE SKILLS

Modelling	Time series & forecasting, missing-data imputation, diffusion models (EDM), JEPA / self-supervised learning, SSM/Mamba, Transformers, LSTM/GRU, GNNs, uncertainty quantification & conformal prediction, XAI
LLMs & agents	Prompt engineering (structured outputs, few-shot, self-consistency, injection guards), RAG (LangChain, Qdrant, hybrid BM25 + dense, reranking), LangGraph, MCP, tool calling, OpenAI API, DeepEval, Gradio, Hugging Face, vLLM
Computer vision	Small-object detection, RT-DETRv2, DINOv2, YOLO, SAHI tiled inference, Grad-CAM
Engineering & MLOps	Python, PyTorch (DDP, FSDP), TensorFlow, Docker, Git, CI/CD, PyTest, MLflow, W&B, OpenTelemetry, Prometheus, ONNX Runtime, Slurm/HPC, SQL, C++

EXPERIENCE

LS2N, Nantes Digital Sciences Laboratory (CNRS UMR 6004) — Nov. 2024 – present

Postdoctoral Researcher, AI for clinical time series and patient digital twins — *Nantes, France*

- Lead the technical architecture of a high-fidelity patient simulator (digital twin under general anaesthesia) for clinical decision support and medical training; designed the temporal neural engine that simulates multivariate intraoperative physiology in real time.
- Built streaming inference pipelines for low-latency prediction, per-patient personalisation and interpretable outputs under clinical constraints (PyTorch, Unity integration, HPC).
- Created three models for joint imputation and prediction on sparse time series (**TF-JEPA, DyFAIP, FASTER**; 3 papers under review), evaluated on MIMIC-III, PhysioNet-2012 and ETT.
- Designed **FASTER-P**, a single recurrent model that handles every missingness type and rate (MCAR/MAR/MNAR, 20–80%), benchmarked on identical masks against BRITS, GRU-D, SAITS, TimesNet and ImputeFormer; extended it into an EDM diffusion imputer.
- Developed **calibrated residual diffusion**, which adds calibrated uncertainty to any imputer in 1–2 network passes: **90.9%** raw coverage of a 90% interval versus 77–79% for standard samplers, and **31–41% lower CRPS** than full diffusion on the same backbone (preliminary).

ICube, University of Strasbourg (CNRS UMR 7357) — Oct. 2020 – Oct. 2024

Ph.D. Researcher, deep temporal networks for irregular ICU trajectories — *Strasbourg, France*

- Designed and open-sourced two dynamic-memory architectures for sparse, irregular health records, **MWTA-LSTM** and **AMITA-LSTM**, reaching **AUROC 0.93** / **AUPRC 0.78** for outcome prediction on eICU (200,000+ admissions, 208 US hospitals); published in *Information Sciences* and *Expert Systems with Applications*.
- Built per-variable attention and frequency weighting that explain each prediction, and validated the explanations with ICU heads of department at Strasbourg University Hospital.
- Built automated ingestion and preprocessing pipelines for MIMIC-III (40,000+ patients) and eICU, with adaptive pooling to limit the effect of extreme values; trained on a Slurm HPC cluster.

ICube, University of Strasbourg — Apr. – Sep. 2020

Master's Research Intern: interpretable anomaly detection (LSTM + shapelets, class activation maps) — *Strasbourg, France*

SELECTED APPLIED PROJECTS

Drone imagery: detecting objects a few pixels wide — 2025 – 2026

RT-DETRv2, DINOv2 ViT-B, YOLO26-P2, SAHI/ASAHI, DBSCAN, Grad-CAM

- Integrated a self-supervised DINOv2 ViT-Base backbone with multi-scale projection (P3–P5) into RT-DETRv2; two-pass inference on 20 MP images (global 3200 × 3200 scan, then dynamic SAHI tiling with Greedy NMM merging).
- Designed a Top-K spatial-attention module that re-injects high-resolution P2/P3 features into the YOLO detection head, with DBSCAN-guided tiling, rare-class oversampling and an IoU-based audit of every missed detection.

LLM agents with hybrid retrieval, tools and observability — 2026

LangGraph, MCP, BM25 + dense, RRF, cross-encoder, Pydantic, DeepEval, OpenTelemetry, React

- Conversational commerce agent that routes each request to document retrieval or to typed tools (order tracking, cart management); hybrid BM25 + dense retrieval fused with Reciprocal Rank Fusion and reranked by a cross-encoder.
- Automated quality gates in CI (PyTest + DeepEval on faithfulness and context relevance) and OpenTelemetry/Prometheus tracing of agent paths, token latency and tool-call reliability.
- LangGraph multi-agent applications (web research, code review, PDF Q&A, debate) with MCP servers and React/Vite front ends.

RAG over PDF documents with page citations 2026

LangChain, Qdrant (Docker), OpenAI text-embedding-3-large, gpt-4o-mini

- Indexing pipeline: PDF loading with page metadata, recursive character splitting (1,000-character chunks, 400 overlap), embeddings stored in a Qdrant vector database; question answering by top-3 similarity search with answers that point to the source page.

Tool-using LLM agent and prompt engineering 2026

OpenAI API (JSON mode), gpt-4o-mini, Gradio, structured outputs, few-shot, self-consistency

- Agent that reasons through explicit START, PLAN, TOOL, REVIEW and OUTPUT steps in JSON, with code-side routing to a weather API and shell tools, a bounded 15-step loop and a Gradio interface that streams every step.
- Prompt design for production use: role and rules, few-shot examples, JSON-schema outputs, grounding with citations, prompt-injection guards and tool allow-lists, self-consistency voting and self-critique, with automatic checks of model replies.

Anomaly detection in industrial sensor networks 2025 – 2026

MTGNN, GNN-LSTM, SWaT, WADI

- Graph-based detectors evaluated on water-treatment and distribution testbeds (51 and 120+ sensors) under cyber-physical attacks.

PUBLICATIONS

[J.1] **Cissoko, M. B. H.**, Castelain, V., et al. (2025). Predicting and Interpreting Healthcare Trajectories from Irregularly Collected Sequential Patient Data Using AMITA. *Information Sciences*, 705. **[Link]**

[J.2] **Cissoko, M. B. H.**, Castelain, V., et al. (2025). Multi-Way Adaptive Time-Aware LSTM for Irregularly Collected Sequential ICU Data. *Expert Systems with Applications*, 261. **[Link]**

[C.1] **Cissoko, M. B. H.**, et al. (2024). Modeling Temporal Dynamics in Irregular ICU Data Using MWTA-LSTM. *EPIA 2024*, Springer LNCS. **[Link]**

[C.2] **Cissoko, M. B. H.**, et al. (2024). Predicting Patient Health Outcomes with AMITA from Irregular Time Series. *ICONIP 2024*, Springer LNCS. **[Link]**

[S.1] **Cissoko, M. B. H.**, Sinoquet, C., et al. FASTER: Frequency-Conditioned Diffusion for Joint Imputation and Prediction in Sparse Time Series. *Expert Systems with Applications* (under review).

[S.2] **Cissoko, M. B. H.**, Sinoquet, C., et al. DyFAIP: End-to-End Joint Imputation and Prediction in Irregular Time Series via Adaptive Frequency Transitions. *Expert Systems with Applications* (under review).

[S.3] **Cissoko, M. B. H.**, Sinoquet, C., et al. TF-JEPA: Time-Frequency Joint Embedding Predictive Architecture for Incomplete Time Series. (under review).

TEACHING & ACADEMIC SERVICE

- **Lecturer, Nantes Université** (Nov. 2024 – present): Master's (LSTM/GRU, Transformers, optimisation) and Bachelor's (statistical learning, evaluation metrics); PyTorch and scikit-learn labs for 60+ engineering students.
- **Lecturer, Vietnam National University** (Dec. 2024 – present): Master's course on sequence models and attention.
- **Reviewer**: NeurIPS, UAI, AISTATS; *Information Sciences* (30+ reviews), *Neurocomputing*, *Environmental Technology & Innovation*.

EDUCATION

Ph.D. in Computer Science, University of Strasbourg (ICube) **2020 – 2024**

M.Sc. Computer Science, Intelligent Systems & Multimedia, Vietnam National University (IFI), Hanoi **2020**

M.Sc. Computer Science, Data Engineering & Business Intelligence, University of La Rochelle **2020**

B.Sc. Computer Science Applied to Business Management, IUG, Bamako **2013**

AWARDS, LANGUAGES & REFERENCES

Awards: Finalist (top 5), Vietnam AI Grand Challenge 2019; TIBCO-IFI Excellence Scholarship 2020; CCNA and MCITP certifications 2017.
Languages: French (native), English (fluent), Vietnamese (basic).
References: Prof. Nicolas Lachiche and Prof. Vincent Castelain (Ph.D. supervisors, University of Strasbourg); Dr. Christine Sinoquet (postdoctoral supervisor, Nantes Université). Contact details on request.